A Computer-Vision-Based Bikeability Score

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***Abstract*—**

***Index Terms*—Bikeability, computer vision, zero-shot classification, vision-language models, CLIP, surface classification, object detection, GPX.**

I. Introduction

II. Research Approach and Methodology

III. Objectives and Constraints

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IV. Ground Detection: Road-Surface Classification

The ridden surface is a primary determinant of cycling comfort and safety: a paved cycleway can be ridden effortlessly, whereas loose gravel or bare soil is strenuous and, on a road bike, potentially hazardous. Ground detection therefore assigns each sampled frame a single surface class and supplies the surface term of the bikeability score. Frames come from a handlebar-mounted camera, sampled at one frame every five seconds; as each frame is dominated by one surface, the task is single-label image classification, not segmentation.

A. Task Definition and Model Selection

Two model families can solve this task without an existing labeled corpus: a classifier *trained on our own labels*, or a *zero-shot* vision-language model (VLM). We avoid a self-trained classifier, as it would need a large, balanced, manually labeled training set, careful regularization on limited data, and complete re-training whenever the class definitions change. A zero-shot VLM of the CLIP family [1] instead classifies an image from its similarity to natural-language class descriptions, so the taxonomy can be adapted by editing text – without task-specific training.

One distinction is often blurred: *linear probing* of self-supervised features (e.g., DINOv2 [2]) is *not* zero-shot – its linear head is fitted on labeled target data and is therefore supervised; it is excluded here and would only qualify as a declared supervised baseline.

B. Surface Classes and Prompts

We distinguish four cycling-relevant classes: **Cycleway** (a dedicated paved bike path), **Road** (a paved road for motor traffic), **Gravel** (loose crushed stone), and **Unpaved** (soil, mud, or field tracks). Each class is described by several English prompts whose normalized text embeddings are averaged (*prompt ensembling*). Since Cycleway and Road share the same smooth-asphalt texture, the prompts encode the *discriminating context* – a narrow separated lane with bicycle markings versus a wide multi-lane road – to push the two embeddings apart. A frame is assigned to the class of highest cosine similarity via a softmax.

C. Evaluation Protocol

We compare nine open CLIP-family models of light to medium size through the unified `open_clip` API [3]: OpenAI CLIP, OpenCLIP (LAION-2B, DataComp-XL) [4], SigLIP [5], MobileCLIP [6], and EVA-CLIP [7]. Both *accuracy* and *latency* matter for on-bike use. For the quantitative evaluation we manually labeled more than 1,700 frames from self-recorded rides; frames without a clear surface were tagged *unclear/other* and excluded. As the classes are imbalanced, we report accuracy, balanced accuracy and macro-F1 with 95% bootstrap confidence intervals and McNemar tests; latency is measured at batch size 1 with warm-up runs (ms/frame and FPS).

Table I
Zero-shot surface-classification benchmark on the self-labeled test set ($>$1,700 frames), sorted by macro-F1. BAcc: balanced accuracy; latency at batch size 1.

Model	**P[M]**	**Acc**	**BAcc**	**mF1**	**ms**	**FPS**
OpenCLIP B/32 (LAION-2B)	151	0.75	**0.72**	**0.58**	115	8.7
SigLIP B/16 (WebLI)	203	0.75	0.62	0.57	259	3.9
CLIP B/16 (OpenAI)	150	**0.78**	0.62	0.57	261	3.8
MobileCLIP-S2	99	0.73	0.70	0.56	304	3.3
CLIP B/32 (OpenAI)	151	0.70	0.60	0.54	93	10.7
EVA02-B/16	150	0.76	0.62	0.54	304	3.3
OpenCLIP B/32 (DataComp-XL)	151	0.75	0.65	0.52	125	8.0
CLIP L/14 (OpenAI)	428	0.64	0.50	0.42	929	1.1

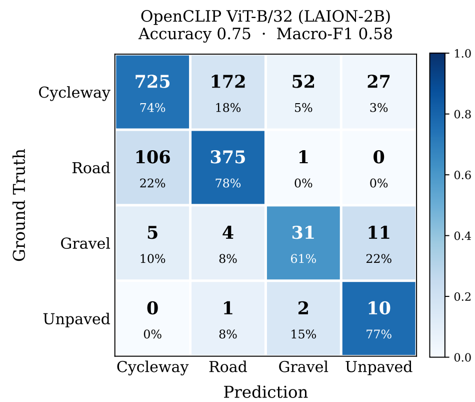

Figure 1. Row-normalized confusion matrix of the best model (OpenCLIP ViT-B/32, LAION-2B) on the labeled test set. Each cell shows the count and the row share; the diagonal is correct. Errors concentrate on the Cycleway/Road pair and the rare Gravel/Unpaved classes.

D. Results

Table I lists the outcome, sorted by macro-F1. The best model, **OpenCLIP ViT-B/32 (LAION-2B)**, reaches macro-F1 0.58 at only 115 ms/frame – the best accuracy-latency trade-off. Two points stand out. First, the highest plain *accuracy* (0.78, CLIP ViT-B/16) does *not* give the highest macro-F1: accuracy rewards the majority classes, whereas macro-F1 weights all four surfaces equally and is the more faithful measure on imbalanced data. Consistently, OpenCLIP ViT-B/32 (LAION-2B) also attains the clearly highest balanced accuracy (0.72 vs. $\leq$ 0.70 for every other model). Second, the 95% bootstrap accuracy intervals of the leading models overlap, so their differences are *not* statistically significant; only CLIP ViT-L/14 (428 M) falls clearly below them – despite being the largest model it is both slowest and weakest (0.42 macro-F1), confirming that capacity alone does not help. As Fig. 1 shows, the residual errors concentrate on the Cycleway/Road pair (both smooth asphalt) and on the rare Gravel and Unpaved classes, which hold only 51 and 13 labeled frames and thus cap the macro-F1 at 0.58 – still sufficient for one of three sub-scores. We therefore adopt OpenCLIP ViT-B/32 (LAION-2B) as the ground detector. As all models share the same input and `open_clip` pipeline, the differences reflect architecture and training data; the CPU latencies are hardware-dependent.

V. Object Detection

VI. Environment Detection

VII. Computation of the Bikeability Score

The bikeability score combines the three sub-components – surface (ground detection), environment (environment detection), and objects (object detection) – into a single value in the interval $[0, 100]$. Every detector emits a fixed-length *vector*; each vector is first mapped to a *sub-score* in $[0, 1]$, and the three sub-scores are then combined by a weighted mean.

A. Detector Outputs as Vectors

The three detectors deliberately use different but fixed output representations, each aligned element-by-element with a stored value vector:

- **Ground** – a one-hot vector $\mathbf{g} \in \{0, 1\}^4$ over the classes (Cycleway, Road, Gravel, Unpaved); exactly one entry is 1.
- **Environment** – a vector of area fractions $\mathbf{f} \in [0, 1]^3$ over (vegetation, water, city).
- **Object** – a vector of counts $\mathbf{n} \in \mathbb{Z}^3_{\geq 0}$ over (bicycle, car, traffic light).

B. Overall Formula

For each evaluated frame, the score A results from the weighted mean of the three sub-scores:

$$A = 100 \cdot \Big(w_g\, S_{\text{ground}} + w_e\, S_{\text{env}} + w_o\, S_{\text{object}}\Big), \tag{1}$$

with the weights

$$w_g = 0.20, \quad w_e = 0.45, \quad w_o = 0.35, \quad w_g + w_e + w_o = 1. \tag{2}$$

The weights are heuristic and follow two considerations. Environment and object exposure jointly capture how pleasant and how safe a stretch feels – the factors a cyclist perceives most directly – and therefore receive the larger share ($w_e + w_o = 0.80$). The surface receives the smallest weight ($w_g = 0.20$): on typical routes it varies little, and ground detection is also the least reliable component (macro-F1 0.58), so a smaller weight limits the influence of its errors. Since all sub-scores lie in $[0, 1]$ and the weights sum to 1, the score is bounded to $[0, 100]$; in practice the extremes 0 and 100 are never reached, as $S_{\text{ground}} \geq 0.1$ and $S_{\text{object}} = \exp(-\max(\rho, 0)) > 0$.

C. Ground Sub-Score

The ground sub-score is the inner product of the one-hot vector $\mathbf{g}$ with the surface quality vector $\mathbf{q}^g = (1.0,\ 0.7,\ 0.3,\ 0.1)$ (Table II):

$$S_{\text{ground}} = \langle \mathbf{g},\, \mathbf{q}^g \rangle = \sum_k g_k\, q^g_k. \tag{3}$$

Because $\mathbf{g}$ is one-hot, this simply selects the quality of the detected surface class.

Table II

Class value vectors for the three sub-scores: surface quality $\mathbf{q}^g$, environment quality $\mathbf{q}^e$, and object weight $\boldsymbol{\lambda}$ (heuristic; a negative value acts as a bonus).

Component	Class	Value
Ground $\mathbf{q}^g$	Cycleway	1.0
	Road	0.7
	Gravel	0.3
	Unpaved	0.1
Environment $\mathbf{q}^e$	Vegetation	1.0
	Water	1.0
	City	0.4
Object weight $\boldsymbol{\lambda}$	Bicycle	-0.2
	Car	$+1.0$
	Traffic light	$+0.3$

D. Environment Sub-Score

The environment sub-score is the fraction-weighted mean of the environment quality vector $\mathbf{q}^e = (1.0,\ 1.0,\ 0.4)$:

$$S_{\text{env}} = \begin{cases} \dfrac{\langle \mathbf{f},\, \mathbf{q}^e \rangle}{\sum_k f_k}, & \text{if } \sum_k f_k > 0, \\ 0.5, & \text{otherwise.} \end{cases} \tag{4}$$

The normalization by $\sum_k f_k$ keeps the sub-score in $[0, 1]$ even when the detected fractions do not cover the whole frame; a neutral value of 0.5 is used if no environment class is detected.

E. Object Sub-Score

The object sub-score maps the counts $\mathbf{n}$ through a *saturation function* using the penalty vector $\boldsymbol{\lambda} = (-0.2,\ 1.0,\ 0.3)$:

$$S_{\text{object}} = \exp\Big(-\max\big(\rho, 0\big)\Big), \qquad \rho = \langle \mathbf{n},\, \boldsymbol{\lambda} \rangle, \tag{5}$$

where ρ is evaluated independently per frame. A positive factor for cars ($+1.0$) and traffic lights ($+0.3$) makes ρ positive and thus lowers the sub-score, whereas bicycles carry a *negative* factor (-0.2). The clamp $\max(\rho, 0)$ makes the score asymmetric: a frame is never rewarded *above* the neutral value 1.0, so bicycles do not inflate the score on their own – they only *offset* the penalty of nearby cars or traffic lights. This encodes the intuition that good cycling conditions are the default and only motorised traffic degrades them. The exponential form additionally caps the influence of outliers, so a single crowded frame cannot dominate the overall score.

F. Route Processing over the Full Ride

The action camera splits one continuous recording into several files (e.g., `DJI_0408`, `DJI_0409`, ...). All parts are processed as a *single* continuous video: the files are ordered chronologically by the number in their name, and a cumulative time offset (the sum of the previous parts' durations) keeps the timeline gap-free across file boundaries. Within each part, one frame is sampled every 5 s by *seeking* directly to the target frame index instead of decoding every frame, which removes almost all of the decoding work. The three detectors are loaded once and evaluated in batches; for every sampled frame, the three output vectors and the resulting score are written to a shared CSV (frame name, continuous time in seconds and `HH:MM:SS`, an optional absolute ISO timestamp, the serialized vectors, and the score).

G. GPX Mapping and Visualization

Because the camera clock is unreliable, the frames are aligned to the GPS track by a single *manual sync point* (by default: recording start = GPX track start). Each frame is then matched to its nearest GPX track point in time (a nearest-neighbor join within a 3 s tolerance – large enough to bridge gaps in the roughly 1 Hz GPS log, yet below the 5 s frame spacing so that each frame maps to a distinct track point), which assigns a latitude/longitude to every score. The *average* bikeability score of the route is the mean $\bar{A} = \frac{1}{M}\sum_{i=1}^{M} A_i$ over all M evaluated frames. Finally, the route is drawn as a sequence of short segments, each colored by the local score on a red–orange–yellow–green scale (Fig. 2). For the example ride shown, the mean score is $\bar{A} \approx 85.5/100$: the track is predominantly green (good cycling conditions) with a few yellow/orange sections.

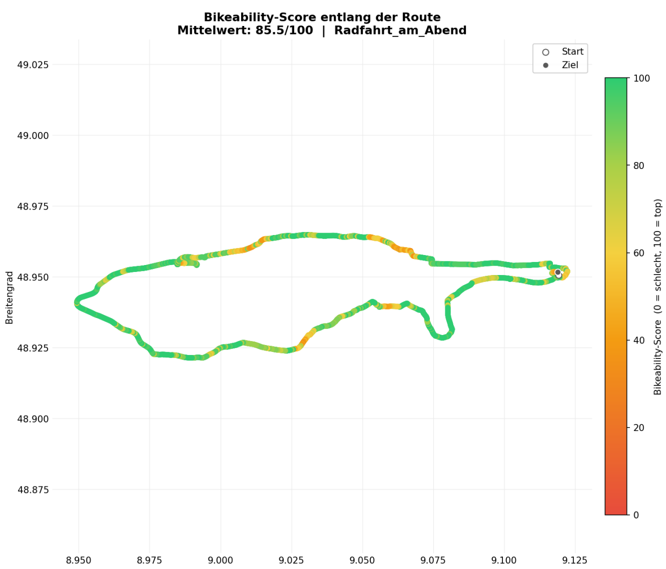

Figure 2. Bikeability score colored along the GPX route (red = low, green = high). Per-frame scores are matched to the nearest GPS track point via a manual sync point; the example ride has a mean score of $\bar{A} \approx 85.5/100$.

VIII. Discussion

IX. Conclusion

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